

# Bintang Dwi Marthen

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## Research Interests

Systems-for-ML, ML-for-systems, operating systems, cloud computing, storage systems, and distributed systems.

## Education

### Harvard University

*PhD in Computer Science*

*Cambridge, Massachusetts*

*Starting Sep 2026*

- Advisor: Juncheng Yang

### Bandung Institute of Technology

*MS in Computer Science*

*Bandung, Indonesia*

*Sep 2025 – Exp. Jul 2026*

- #1 Engineering School in Indonesia
- GPA: 3.94/4.0
- Advisor: Achmad Imam Kistijantoro
- **Coursework:** Advanced Distributed Systems, Advanced Operating System, Cloud Computing Architecture, Deep Learning

### Bandung Institute of Technology

*BS in Computer Science*

*Bandung, Indonesia*

*Aug 2021 – Jul 2025*

- #1 Engineering School in Indonesia
- GPA: 3.89/4.0 (Major GPA: 3.86/4.0)
- Advisor: Achmad Imam Kistijantoro
- **Coursework:** Computer Organization and Architecture, Operating System, Computer Networks, Parallel and Distributed Systems, Distributed Application Architecture, Machine Learning, Database Management

## Publications

### Learning-Augmented Heuristics: Simple, yet Smart, Robust and Interpretable Cache Eviction

OSDI 2026

Haocheng Xia, William Nixon, **Bintang Dwi Marthen**, Pranav Bhandari, Juncheng Yang

### Clock2Q+: A Simple and Efficient Replacement Algorithm for Metadata Cache in VMware vSAN

VLDB 2026

Yiyan Zhai, **Bintang Dwi Marthen**, Sarath Balivada, Vamsi Sudhakar Bojji, Eric Knauft, Jitender Rohilla, Jiaqi Zuo, Quanxing Liu, Maxime Austruy, Wenguang Wang, Juncheng Yang

## Research Experience

### Serverless Platform for Development Workload

*International Research Collaboration*

*Harvard University*

*Apr 2025 – Present*

- Collaborating with Assistant Professor Juncheng Yang (Harvard University) on designing a developer-friendly serverless platform to eliminate idle resource waste inherent in VM-based development workflows.
- Analyzed fine-grained execution traces from thousands of workloads to systematically characterize latency and resource inefficiencies in emerging serverless environments.
- Discovered that container image construction dominates startup overhead, accounting for up to 82% of total initialization time.
- Initial prototyping of optimized base image synthesis achieved up to 99.6% reduction in cold-start initialization time.

### Feasibility of Static Parameterization in Adaptive Caches

*International Research Collaboration*

*Harvard University*

*Mar 2025 – Present*

- Collaborating with Assistant Professor Juncheng Yang (Harvard University) on re-evaluating how much adaptivity is actually needed in cache management.

- Demonstrated through extensive simulation that statically parameterized cache policies can outperform their adaptive counterparts by up to 20% in hit ratio.
- Uncovered that adaptive cache performance depends on the granularity match between the adaptation mechanism and the workload's phase behavior; mismatches in either direction lead to performance loss.
- Investigating phase-change and workload drift detection mechanisms to enable coarse-grained, stability-aware adaptivity with minimal runtime and tuning overhead.

### System and AI Research (SYAIR) Training Program

University of Chicago

Research Training Program

Jan 2025 – Jul 2025

- Selected as one of the top 50 students in Indonesia for a research training program led by Prof. Haryadi S. Gunawi (University of Chicago).
- Reproduced the *Clock-PRO* cache replacement algorithm as a Chameleon Trovi artifact.
- Studied over 30 research papers from top systems conferences (*SOSP*, *OSDI*, *NSDI*, *FAST*, and others).

### Sentiment Analysis Using LLMs for Twitter Data

Bandung, Indonesia

Undergraduate Researcher

Oct 2024 – Dec 2024

- Collaborating with Ayu Purwianti (Bandung Institute of Technology) and Achmad Imam Kistijantoro (Bandung Institute of Technology) on leveraging large language models for sentiment analysis of Indonesian Twitter posts.
- Fine-tuned pre-trained transformer models and evaluated domain adaptation strategies for low-resource Indonesian text.
- Maintained and optimized NVIDIA DGX servers for large-scale model training.
- Built an internal web interface to streamline prompt testing and result comparison across multiple LLMs.
- Optimized source code, accelerating the experimentation process by approximately 10×

## Work Experience

### Research Intern

Bandung, Indonesia

Formulatrix

Jun 2024 – Sep 2024

- Designed and implemented a proof of concept for the automatic map discovery feature of the ROVER product, reduced configuration from three weeks to three days for non-technical users
- Introduced Simultaneous Localization and Mapping (SLAM) using a monocular front-facing camera, reducing camera count from two to one and cutting manufacturing costs by approximately 4.4%
- Modified OpenCV ArUco to support STag markers in singular and composite configurations, improving marker flexibility for prototype testing

## Highlighted Projects

### Simple x86 32-bit Operating System

[github repo](#) 

- Developed a basic operating system kernel from scratch, with support for interrupts, keyboard input, and FAT32 file system management
- Implemented key file system operations such as `ls`, `cd`, `mkdir`, `whereis`, `cat`, `rm`, `cp`, and `mv`, providing full CRUD functionality

### Clock-PRO Cache and CACHEUS Fix

[github repo](#) 

- Implemented Clock-PRO cache algorithm on libCacheSim, a high-performance library for cache simulation
- Debugged and fixed faulty CACHEUS implementation

## Technical Skills

**Languages:** C++, C, Python, Go, C#, SQL, JavaScript, Kotlin, Java

**Machine Learning:** vLLM, Ollama, Transformers, PyTorch, TensorFlow, Scikit-learn, Pandas

**Databases and Systems:** SQLite, MySQL, PostgreSQL, MongoDB, Redis, CouchDB

**Systems:** RabbitMQ, FEMU, libCacheSim

**Cloud Computing:** Google Cloud, AWS, Chameleon Cloud, CloudLab, Amazon S3

**DevOps:** Kubernetes, GitHub CI/CD, containerd, firecracker microVM, Docker

**Misc:** LaTeX, Make, CMake